

Supplemental material 4:

Checklist: OPM-MEG operator for infancy research

Please note that **this checklist** has been developed for HEDscan system, FieldLine Medical, Boulder, CO, US, and **may need to be adapted to your system / lab needs**.

1. Arrive 30 minutes before the session
2. Make sure the MSR & the lab space is tidy
3. Switch on the computers
 - a. Keep the [stimulus delivery script] script ready (do a trial run)
 - b. Switch on the [data acquisition] computer
 - c. **Do not** switch on the chassis/sensors yet
 - d. Set up for the video recording
4. When instructed by the session lead that it's a good time (this will typically happens while the consents forms are being signed, and the family is being prepared for entering the MSR):
 - a. Start Chassis 1 followed by Chassis 2
 - b. Start the sensors and switch off the red/error sensors
 - c. Initialize the sensors while keeping the MSR door is closed (this is especially done when it is the first reading of the day as sometimes this process might take longer when done at first)
 - d. As soon as initialization is done, **switch off** the sensors
5. Going into the MSR
 - a. Check with the lead experimenter if lights are to be kept on or off for the baby to relax.
 - b. Bring the camera and take the pictures to document the baby's position within the helmet.
 - c. Start the video recording & make sure the sound is being recorded.
 - d. **Start the sensors for the recording:** after the baby is settled on the bed, start the sensors and switch off the bad ones. Consult with the lead experimenter to ensure the right timing.
 - e. If needed, assist with taking photographs & sliding the platform inside the helmet.
6. When leaving the MSR:
 - a. Take away glasses, camera, and any other metallic things.
 - b. Close the MSR door.
 - c. Initialize the sensors.
 - d. While the sensors are initializing, enter the folder, file name so that it is ready to start recording right away.
 - e. **Start the recording** followed by the experiment script (use the interphone to confirm with the lead experimenter if okay to start the stimuli presentation)
7. During the scanning
 - a. Monitor the stimulus presentation script.

- b. [study specific instructions]
 - c. Watch out for if the experimenter requests a break, if break taken then stop the recording and start a new one when resumed
 - d. Monitor sensors via FFT and if any of them is wild write down
- 8. After the task is completed, perform 3D scans:
 - a. While the baby is still inside the helmet, do a quick frontal scan
 - b. Then as per the situation, do another outside helmet scan
- 9. After the family leaves the MSR
 - a. Measure and write down the temperature with:
 - i. thermocouples
 - ii. thermal camera
 - b. Record an empty room with localisation (for 5 min)
 - c. Complete the participant log
 - d. Upload the data (OPM and 3D scan) on [lab specific instructions]

Important note: If the family takes a break, switch off the sensor upon consultation with the lead experimenter.